



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

<Name>

<Date>



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
- Summary of all results

Introduction

- Project background and context
- Problems you want to find answers

Section 1

Methodology

Methodology

Executive Summary

Data collection methodology:

- Describe how data was collected

Perform data wrangling

- Describe how data was processed

Perform exploratory data analysis (EDA) using visualization and SQL

- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
- How to build, tune, evaluate classification models
 -

Data Collection

Describe how data sets were collected.

You need to present your data collection process use key phrases and flowcharts

Data Collection – SpaceX API

- GitHub URL:
<https://github.com/tarunkrjaishwal/IBM-Data-Science>

Place your flowchart of SpaceX API calls here

Data Collection - Scraping

- GitHub URL:
<https://github.com/tarunkrjaiswal/IBM-Data-Science>

Place your flowchart of web scraping here

Data Wrangling

GitHub URL: <https://github.com/tarunkrjaiswal/IBM-Data-Science>

EDA with Data Visualization

GitHub URL: <https://github.com/tarunkrjaiswal/IBM-Data-Science>

EDA with SQL

GitHub URL: <https://github.com/tarunkrjaiswal/IBM-Data-Science>

Build an Interactive Map with Folium

GitHub URL: <https://github.com/tarunkrjaiswal/IBM-Data-Science>

Build a Dashboard with Plotly Dash

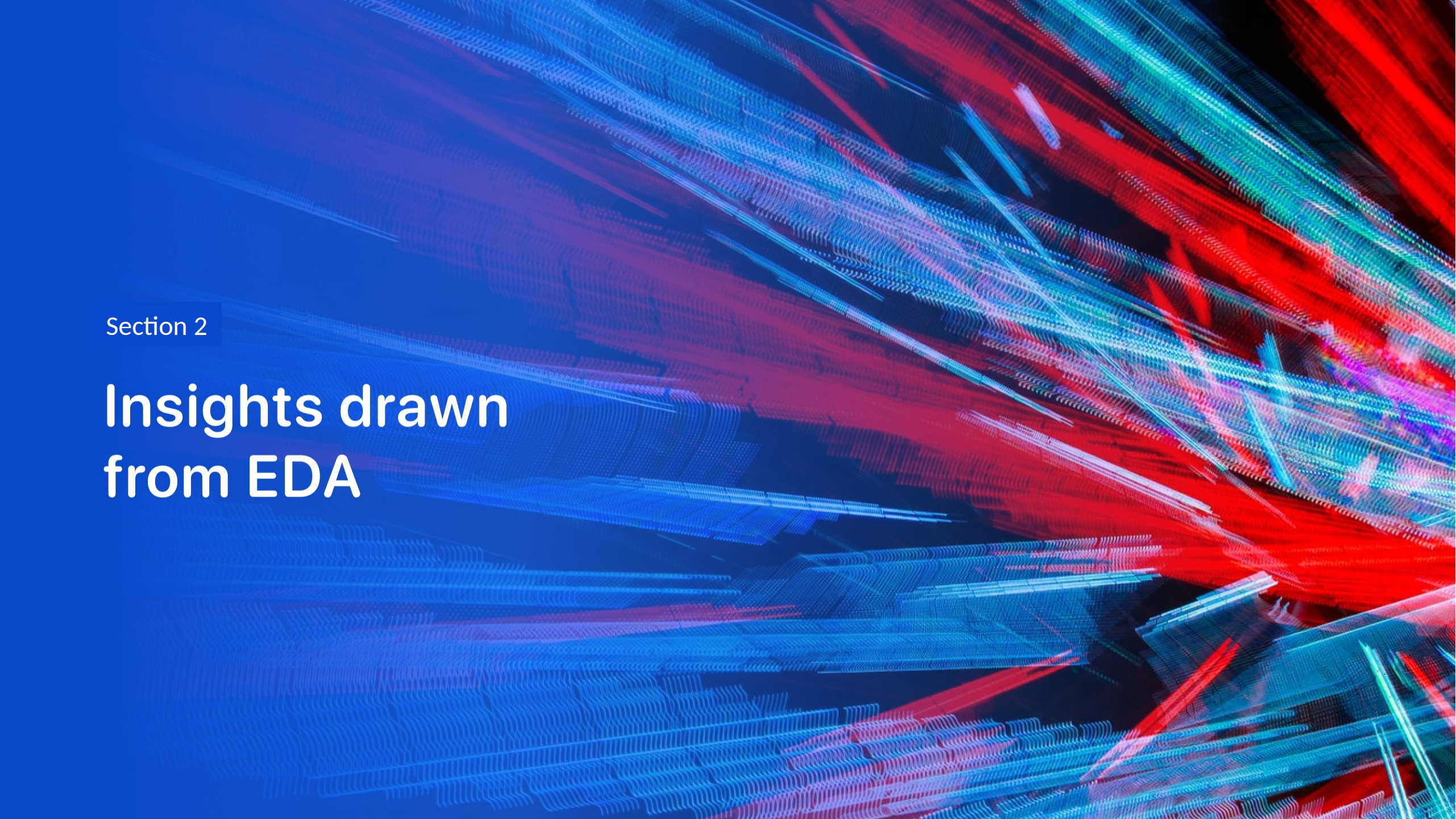
GitHub URL: <https://github.com/tarunkrjaiswal/IBM-Data-Science>

Predictive Analysis (Classification)

GitHub URL: <https://github.com/tarunkrjaiswal/IBM-Data-Science>

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

- Show a scatter plot of Flight Number vs. Launch Site
- Observation: As flight number increases, the success rate also generally improves. Most launches are concentrated at CCAFS and KSC launch sites.

Payload vs. Launch Site

- Show a scatter plot of Payload vs. Launch Site
- Observation: Higher payload masses (e.g. > 10,000 kg) do not negatively impact launch success, and are frequently launched from KSC LC 39A and CCAFS SLC 40.

Success Rate vs. Orbit Type

- Show a bar chart for the success rate of each orbit type
- Observation: Orbits like ES-L1, GEO, HEO, and SSO have a 100% success rate, while SO has a 0% success rate.

Flight Number vs. Orbit Type

- Show a scatter point of Flight number vs. Orbit type
- Observation: Higher flight numbers are correlated with successful insertions into various orbits like VLEO.

Payload vs. Orbit Type

- Show a scatter point of payload vs. orbit type
- Observation: VLEO orbit is heavily used for payloads around 15,000 kg, predominantly successfully.

Launch Success Yearly Trend

- Show a line chart of yearly average success rate
- Observation: The overall launch success rate has been consistently improving over time, climbing sharply from 2013 and maintaining high levels through 2020.

All Launch Site Names

Find the names of the unique launch sites

Query Result: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E. These are the primary launch facilities for SpaceX Falcon 9.

Launch Site Names Begin with 'CCA'

Find 5 records where launch sites begin with `CCA`

Query Result: Successfully retrieved 5 launch records from sites beginning with 'CCA' (CCAFS LC-40).

Total Payload Mass

Calculate the total payload carried by boosters from NASA

Query Result: Total Payload Mass for NASA (CRS) is 45,596 kg.

Average Payload Mass by F9 v1.1

Calculate the average payload mass carried by booster version F9 v1.1

Query Result: The average payload mass for booster version F9 v1.1 is 2,928 kg.

First Successful Ground Landing Date

Find the dates of the first successful landing outcome on ground pad

Query Result: The first successful ground pad landing was on 2015-12-22.

Successful Drone Ship Landing with Payload between 4000 and 6000

List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

Query Result: Boosters F9 FT B1022, B1026, B1021.2, and B1031.2 successfully landed on drone ships with payload between 4000 and 6000 kg.

Total Number of Successful and Failure Mission Outcomes

Calculate the total number of successful and failure mission outcomes

Query Result: 73 Successful missions, 27 Failed missions (including various modes of failure).

Boosters Carried Maximum Payload

List the names of the booster which have carried the maximum payload mass

Query Result: Boosters B1048.4, B1049.4, B1051.3, B1056.4 carried the maximum payload of 15,600 kg.

2015 Launch Records

List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

Query Result: In 2015, F9 v1.1 B1012 and B1015 failed drone ship landings from CCAFS LC-40.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

Query Result: Success (drone ship) and Success (ground pad) were the most frequent outcomes between those dates.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky and a view of the Earth's surface, which is covered in a dense network of city lights and clouds. The lights are concentrated in the lower right portion of the image, while the upper left shows a clear blue sky.

Section 3

Launch Sites Proximities Analysis

<Folium Map Screenshot 1>

Launch Sites on Global Map

Explore the generated folium map and make a proper screenshot to include all launch sites' location markers on a global map

Observation: All launch sites are located near the coastline to ensure safety over the ocean during launches.

<Folium Map Screenshot 2>

Launch Outcomes on Global Map

Explore the folium map and make a proper screenshot to show the color-labeled launch outcomes on the map

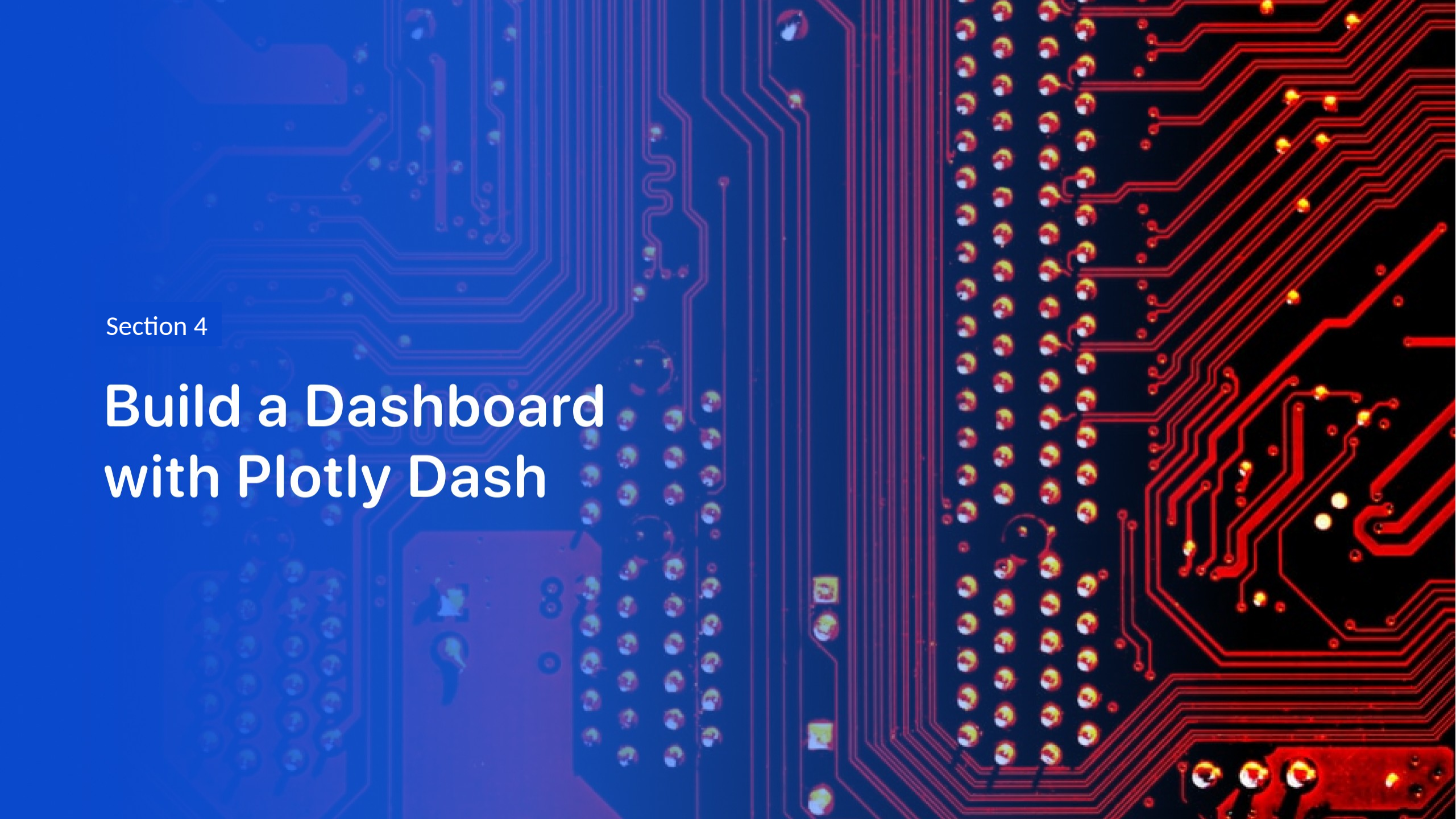
Observation: Successful landings (green) and failures (red) show that KSC LC-39A has the highest concentration of successful outcomes.

<Folium Map Screenshot 3>

Launch Site Proximities

Explore the generated folium map and show the screenshot of a selected launch site to its proximities such as railway, highway, coastline, with distance calculated and displayed

Observation: Launch sites are strategically placed near railways and highways for transportation logistics, and far from densely populated areas.



Section 4

Build a Dashboard with Plotly Dash

<Dashboard Screenshot 1>

Launch Success Pie Chart (All Sites)

Show the screenshot of launch success count for all sites, in a piechart

Observation: KSC LC-39A contributed to the largest share of successful launches.

<Dashboard Screenshot 2>

Launch Success Pie Chart (KSC LC-39A)

Show the screenshot of the piechart for the launch site with highest launch success ratio

Observation: For KSC LC-39A, the vast majority of launches were successful.

<Dashboard Screenshot 3>

Payload vs Launch Outcome Scatter Plot

Show screenshots of Payload vs. Launch Outcome scatter plot for all sites, with different payload selected in the range slider

Observation: Payloads between 2000-4000 kg have mixed outcomes, but higher payloads tend to be successfully launched more consistently., such as which payload range or booster version have the largest success rate, etc.

Section 5

Predictive Analysis (Classification)

Classification Accuracy

- Visualize the built model accuracy for all built classification models, in a bar chart
- The Decision Tree model achieved the highest test classification accuracy (83.3%), matching the SVM model.

Confusion Matrix

- The confusion matrix shows the model correctly predicted 12 True Positives (successful landings) and 3 False Positives (predicted success, but actually failed). It effectively distinguishes classes but struggles slightly with false positives.

Conclusions

1. Launch success rates have improved significantly over time, particularly after 2013.
2. Orbits like ES-L1, GEO, HEO, and SSO have 100% success rates.
3. KSC LC-39A is the most successful launch site for Falcon 9.
4. Machine learning classification models can accurately predict landing outcomes, with Decision Tree classifiers providing robust accuracy.

Appendix

Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

